

Questions on the matching-vector TreeEval construction

Notes for the authors | 1 August 2026

I wrote exact checkers for the matching-vector and selector identities while working through the paper. The main question that came out of this is about the binary representation of the catalytic registers. I have included two smaller parameter observations for context.

1. Can the catalytic registers be packed jointly?

HPR Remark 2.3 validates Z_m coordinates separately and gives $O(d \log(dm))$ binary catalytic space. It seems possible to avoid the extra $\log d$ factor by packing all $D=\Theta(d)$ coordinates into one integer.

$$M=m^D, \quad b=\lceil \log_2 M \rceil, \quad y=qM+x, \quad q \in \{0,1\}, \quad 0 \leq x < M$$

Interpret the arbitrary initial b catalyst bits as y in $[0, 2^b)$. Since $2^b < 2M$, save the one-bit quotient q , subtract qM in place, and read $x = \sum_i x_i m^i$ as the vector in Z_m^D . Once the ring algorithm has restored x , adding qM restores the original binary tape exactly.

Packed-coordinate access

Coordinate i is $\text{floor}(x/m^i) \bmod m$. If it is changed by a , retain the old $O(\log m)$ -bit digit and add the signed integer

$$((x_i + a \bmod m) - x_i) m^i$$

to the packed tape in a low-to-high carry or borrow pass. Bits of powers, products, and quotients can be generated in logspace using the uniform TC0 arithmetic of Hesse-Allender-Barrington. Reads and inner products stream over the packed digits. HPR use a constant number of registers, so their coordinates can be concatenated and register swaps handled by a constant-size logical permutation.

This appears to use $\lceil D \log_2 m \rceil$ catalyst bits and $O(\log m + \log(D \log m))$ additional work bits, with polynomial overhead. In Theorem 3.1 it would replace $O(d \log(dm))$ by $O(d \log m)$ without changing the stated free-space or polynomial-time bounds.

BCKLS remark after Lemma 15 that stronger compression of the high-order bits should reduce their non-power-of-two simulation to linear catalyst size. I do not view the target as new; the question is whether this state map is a valid realization for HPR's product-ring registers, or whether packed coordinate access violates a constraint of the machine model.

Checks

I checked the state map in an integer implementation and a separate destructive bit-tape implementation. For $m=3, D=4$, both exhaust all 128 initial seven-bit tapes, every coordinate, and every modular update: 1,536 update/inverse transitions in each implementation. The power-of-two boundary is checked separately.

2. The fixed-modulus dimension boundary

Let $N=2^{\ell}$. For constant m, t , using Theorem 3.1 as an ordinary clean-space algorithm would require $d=O(\ell)$ to reach $O(\log n)$ total space. Bhowmick-Dvir-Lovett Theorem 2, together with the now-proved bounded-torsion PFR theorem of Gowers-Green-Manners-Tao, gives for fixed m

$$MV(m, d) \leq \exp(c_m d / \log d).$$

Thus $N=2^{\ell}$ forces $d=\Omega_m(\ell \log \ell)$. This is consistent with the $d=O(\ell \log \ell)$ target in HPR Remark 3.4, which would give polynomial-time $O(\log n \log \log n)$ space, but it rules out reaching L through a fixed-modulus explicit family alone.

Question. My reading is that the remaining L route must use the vectors through a succinct representation rather than materializing all d coordinates. Is that also how you view the boundary?

3. A characteristic-local selector variant

There is a small simplification if the update is projected to one CRT component p_k . At that component, use the shifted product itself instead of selecting one of its monomials. Its signed mixed difference is $xy - (x+1)y - x(y+1) + (x+1)(y+1) = 1$.

For $i \neq k$, keep HPR's (α_i, β_i) filters, normalize by their coefficients, and multiply by the CRT idempotent for p_k . Lemma 3.8 can be run modulo m/p_k , so the baseline inner products stored across its four oracle calls also omit the p_k component. The resulting one-level selector is one at the target and zero elsewhere while suspending $O(t + \log(m/p_k))$ bits rather than $O(\log m)$.

This identity is confined to its output characteristic. A fixed weight $\phi_i: F_p \rightarrow F_q$ cannot provide the same mixed difference for distinct primes: summing over x in F_p makes the left side telescope to zero while the right side is $p \neq 0$ in F_q . The positive identity passes 7,200 direct and 7,200 independently factorized checks over the Z_{15} companion family.

I do not see a way to compose the projected updates recursively without recovering the complementary state, so this is only a one-level observation. **Question.** Could it still be useful for an asymmetric modulus, or is this limitation already captured by the CIR formulation?

References

- [1] A. Henzinger, E. Pyne, S. Ragavan. *Catalytic Tree Evaluation From Matching Vectors*. ECCC TR26-022, 2026.
- [2] A. Bhowmick, Z. Dvir, S. Lovett. *New Bounds for Matching Vector Families*. SIAM J. Comput. 43(5), 2014.
- [3] W. T. Gowers, B. Green, F. Manners, T. Tao. *Marton's Conjecture in Abelian Groups with Bounded Torsion*. 2024/2025.
- [4] H. Buhrman, R. Cleve, M. Koucky, B. Loff, F. Speelman. *Computing with a Full Memory: Catalytic Space*. STOC 2014.
- [5] W. Hesse, E. Allender, D. A. M. Barrington. *Uniform Constant-Depth Threshold Circuits for Division and Iterated Multiplication*. JCSS 65(4), 2002.

Implementation and full derivations are available if useful. Withdrawn ECCC TR26-044 is not used.